

Module 1: Software Processes & Agile

Waterfall, V-Model, Prototyping, Boehm's Spiral 4 Quadrants & Scrum Ceremonies

Module I: Software Process Models & Agile Methodologies

1. The Software Crisis, Engineering Principles & The Dual Nature of Software
2. Generic Software Process Framework (Communication, Planning, Modeling, Construction, Deployment)
3. Classical Waterfall Model: Sequential Phases, Feedback Loops & Major Inherent Flaws
4. The V-Model: Verification vs. Validation Cross-Phase Association
5. Prototyping Paradigm: Throwaway vs. Evolutionary Prototyping Mechanics
6. Boehm's Spiral Model: 4 Quadrants & Explicit Risk-Driven Iterations
7. The Agile Manifesto: 4 Core Values & 12 Guiding Principles
8. The Scrum Framework: Roles (PO, Scrum Master, Team) & Ceremonies
9. Scrum Artifacts: Product Backlog, Sprint Backlog & Sprint Burndown Charts
10. Extreme Programming (XP): Pair Programming, Test-Driven Development (TDD) & CI
11. Process Model Selection Decision Matrix for Diverse Engineering Constraints
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 – 4: Traditional Prescriptive Process Models

A **Software Process Model** is an abstract representation of the lifecycle activities, actions, tasks, milestones, and work products required to build high-quality software.

Process Model	Core Operational Philosophy	Ideal Project Characteristics
1. Classical Waterfall	Strictly linear sequential progression. Downstream phases only begin once upstream deliverables are frozen.	Requirements are 100% clear, stable, and well-understood; safety-critical systems with rigid standards.
2. V-Model	Pairs each development phase on the left descending arm with its corresponding testing verification phase on the right ascending arm.	High-reliability systems where test plan generation occurs early in parallel with requirements.
3. Prototyping	Builds a quick mock-up interface to elicit, clarify, and validate ambiguous user requirements.	High user interaction systems where clients cannot articulate exact requirements upfront.
4. Boehm's Spiral Model	Risk-driven iterative model combining iterative prototyping with Waterfall systematic control. Evaluates risk in 4 quadrants per cycle.	Large-scale, expensive, high-risk enterprise systems with emerging requirements.

Topic 6: Boehm's Spiral Model (4 Quadrants of Iteration)

The 4 Quadrants of the Spiral Model Cycle

1. **Quadrant 1: Determine Objectives, Alternatives & Constraints:** Identify specific phase goals, alternative architectural options, and operational constraints (cost, schedule).
2. **Quadrant 2: Identify & Resolve Risks (Risk Analysis):** Detailed risk evaluation; create simulations, benchmarks, or prototypes to mitigate technical and project risks.
3. **Quadrant 3: Develop & Verify Next-Level Product:** Engineering design, coding, and verification testing of the release increment.
4. **Quadrant 4: Review & Plan Next Phase:** Customer evaluation of the increment, review progress, and plan the next spiral loop.

Topic 7 – 9: Agile Methodologies & The Scrum Framework

The 4 Values of the Agile Manifesto (2001): 1. **Individuals and interactions** over processes and tools. 2. **Working software** over comprehensive documentation. 3. **Customer collaboration** over contract negotiation. 4. **Responding to change** over following a rigid plan.

Scrum Component	Definition & Purpose	Key Responsibility / Invariant
Product Owner (PO)	Represents business stakeholders and customers; defines vision.	Owns and prioritizes the Product Backlog.
Scrum Master	Servant-leader; removes organizational impediments and ensures adherence to Scrum.	Facilitates Daily Standups, Sprint Planning, and Retrospectives.
Sprint Backlog	Subset of Product Backlog user stories selected for delivery in current 2–4 week Sprint.	Estimated by developers using Story Points (Planning Poker).
Burndown Chart	Visual graph plotting remaining work (story points) against time (days of the sprint).	Tracks velocity and predicts whether sprint goals will be met on time.

Top BIT Mesra Exam Questions & Answers (Module I)

Q1. Compare the Waterfall Model and the Spiral Model across 4 engineering parameters. (8 Marks)

1. **Risk Management:** Waterfall has zero explicit risk management mechanisms; Spiral is fundamentally driven by continuous risk analysis in every cycle.
2. **Handling Changing Requirements:** Waterfall resists changes after requirements freeze; Spiral accommodates evolving requirements in subsequent spiral loops.
3. **Customer Involvement:** In Waterfall, customer interaction happens only at the beginning (requirements) and end (delivery); in Spiral, customer evaluates prototypes after every loop.
4. **Project Cost & Complexity:** Waterfall is suitable for small, simple projects; Spiral is designed for large-scale, complex, mission-critical systems.